REGULAR EXPRESSIONS AND CONVERSIONS

IF 2124 TEORI BAHASA FORMAL OTOMATA

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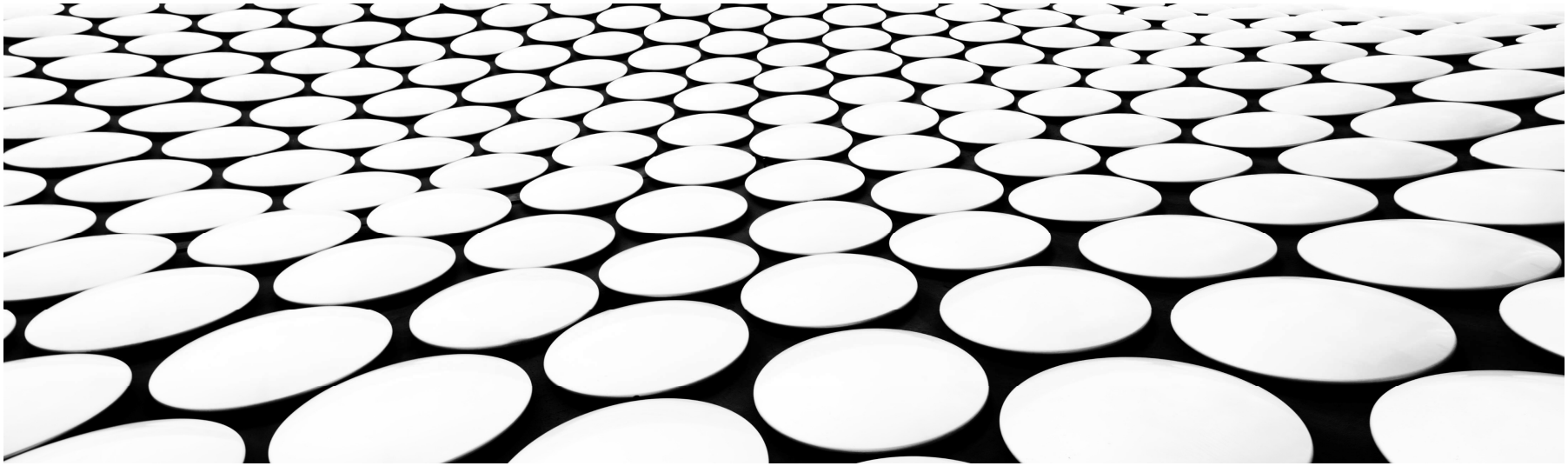


Regular expressions

A FA (NFA or DFA) is a "blueprint" for contructing a machine recognizing a regular language.

A *regular expression* is a "user-friendly," declarative way of describing a regular language.

$10^* = \{1, 10, 100, 1000, \cdots$

$+$

$01^* = \{0, 01, 011, 0111, \cdots$

Example: $\mathbf{01^* + 10^*}$ $= \{0, 1, 01, 10, 100, 011, \cdots$ dst $\}$.

Regular expressions are used in e.g.

1. UNIX `grep` command

2. UNIX Lex (Lexical analyzer generator) and Flex (Fast Lex) tools.

Operations on languages

Union:

$L \cup M = \{w : w \in L \text{ or } w \in M\}$

Concatenation:

$L.M = \{w : w = xy, x \in L, y \in M\}$

Powers:

$L^0 = \{\epsilon\},\ L^1 = L,\ L^{k+1} = L.L^k$

Kleene Closure:

$L^* = \bigcup_{i=0}^{\infty} L^i$ $\{\varepsilon, L, L^2, L^3, \ldots\}$

Question: What are $\emptyset^0$, $\emptyset^i$, and $\emptyset^*$ $\emptyset^0 = \{\varepsilon\}$ $\emptyset^i = \{\}^{i \neq 0}$ $\emptyset^* = \{\varepsilon\}$

Building regex's

Inductive definition of regex's:

Basis: $\boldsymbol{\epsilon}$ is a regex and $\boldsymbol{\emptyset}$ is a regex.
$L(\boldsymbol{\epsilon}) = \{\epsilon\}$, and $L(\boldsymbol{\emptyset}) = \emptyset$.

If $a \in \Sigma$, then $\boldsymbol{a}$ is a regex.
$L(\boldsymbol{a}) = \{a\}$.

Induction:

If E is a regex's, then (E) is a regex.
$L((E)) = L(E)$.

If E and F are regex's, then $E + F$ is a regex.
$L(E + F) = L(E) \cup L(F)$.

If E and F are regex's, then $E.F$ is a regex.
$L(E.F) = L(E).L(F)$.

If E is a regex's, then $E^\star$ is a regex.
$L(E^\star) = (L(E))^*$.

$W \subset \{0,1\}^* = \{\varepsilon, 0, 1, 00 \ldots$

Example: Regex for

$L = \{w \in \{0,1\}^* : \text{ 0 and 1 alternate in } w\}$

$(01)^* = \{\varepsilon, 01, 0101, 010101, \ldots$
$(10)^* = \{\varepsilon, 10, 1010, 101010, \ldots$
$0(10)^* = \{0, 010, 01010, \ldots$
$1(01)^* = \{1, 101, 10101, \ldots$

$$(01)^* + (10)^* + 0(10)^* + 1(01)^*$$

or, equivalently,

$$(\epsilon + 1)(\underset{\varepsilon}{01})^*(\epsilon + 0)$$

$\varepsilon \Rightarrow \{\varepsilon, 0, 1, 01, 10, 1010, 010, \ldots$

$\{\ \}$

Order of precedence for operators:

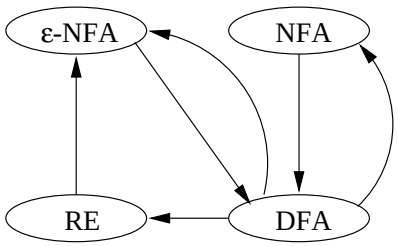
1. Star
2. Dot 01^*
3. Plus $+$, บวก.

$\emptyset = \{\ \} \neq \{\varepsilon\}$
$\emptyset^0 = \{\varepsilon\}$

Example: $01^* + 1$ is grouped $(0(1)^*) + 1$

Equivalence of FA's and regex's

We have already shown that DFA's, NFA's, and ϵ-NFA's all are equivalent.



To show FA's equivalent to regex's we need to establish that

1. For every DFA A we can find (construct, in this case) a regex R, s.t. $L(R) = L(A)$.

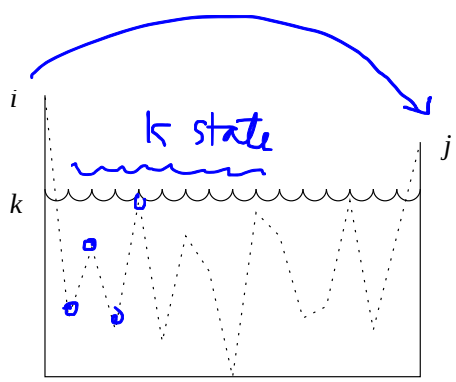
2. For every regex R there is a ϵ-NFA A, s.t. $L(A) = L(R)$.

Theorem 3.4: For every DFA $A = (Q, \Sigma, \delta, q_0, F)$ there is a regex R, s.t. $L(R) = L(A)$.

Proof: Let the states of A be $\{1, 2, \ldots, n\}$, with 1 being the start state.

$R_{ij}^{(k)}$ = Berangkat dari state i ke j melalui k state antara / melewati

• Let $R_{ij}^{(k)}$ be a regex describing the set of labels of all paths in A from state i to state j going through intermediate states $\{1, \ldots, k\}$ only.



$R_{ij}^{(k)}$ will be defined inductively. Note that

$$L\left(\bigoplus_{j\in F} R_{1j}{}^{(n)}\right) = L(A)$$

$R_{1j}^{(n)}$ $j \in F$

Basis: $k = 0$, i.e. no intermediate states.

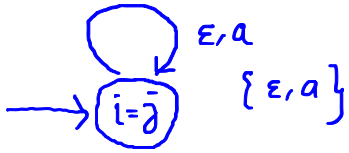
- *Case 1:* $i \neq j$

$k = 0$

$$\Longrightarrow \boxed{R_{ij}^{(0)} = \bigoplus_{\{a\in\Sigma:\delta(i,a)=j\}} a}$$

$R_{ij}^{(0)} = \{a\}$

- *Case 2:* $i = j$



$\{\varepsilon, a\}$

$$\Longrightarrow \boxed{R_{ii}^{(0)} = \left(\bigoplus_{\{a\in\Sigma:\delta(i,a)=i\}} a\right) + \epsilon}$$

Induction:

$$R_{ij}^{(k)}$$

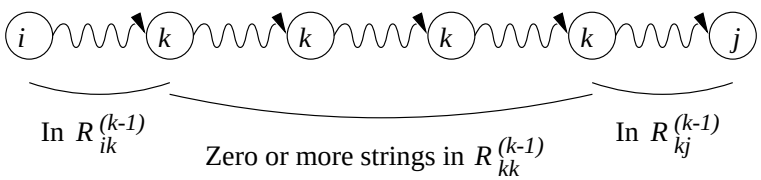
$$=$$

$$R_{ij}^{(k-1)}$$

$$+$$

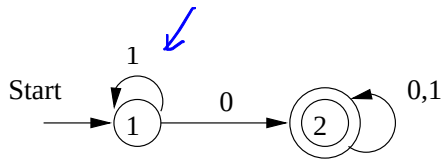
$$R_{ik}^{(k-1)}\left(R_{kk}^{(k-1)}\right)^{*}R_{kj}^{(k-1)}$$

$$\boxed{R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} R_{kk}^{(k-1)} R_{kj}^{(k-1)}}$$



Example: Let's find R for A, where

$L(A) = \{x0y : x \in \{1\}^* \text{ and } y \in \{0,1\}^*\}$



$R_{12}^{(2)} = \{ w \mid \ldots \} = L(A)$

$k = 0$

$R_{11}^{(0)}$	$\epsilon + 1$
$R_{12}^{(0)}$	0
$R_{21}^{(0)}$	$\emptyset$
$R_{22}^{(0)}$	$\epsilon + 0 + 1$

$(\varepsilon + R)^* = \{\varepsilon,$

$R = 0, 1 \quad 01$

$R = 01 \qquad R^* = (01)^*$

$= \{\varepsilon, 01, (01)(01), (01)(01)(01), \ldots$

We will need the following *simplification rules:*

- $(\overset{\varepsilon}{\epsilon} + R)^* = R^*$
- $R + RS^* = RS^*$
- $\emptyset R = R\emptyset = \emptyset$ (Annihilation)
- $\emptyset + R = R + \emptyset = R$ (Identity)

$\boxed{RR^* = RR^*}$

$R_{11}^{(0)}$	$\epsilon + 1$
$R_{12}^{(0)}$	0
$R_{21}^{(0)}$	$\emptyset$
$R_{22}^{(0)}$	$\epsilon + 0 + 1$

$$R_{ij}^{(1)} = R_{ij}^{(0)} + R_{i1}^{(0)} \left(R_{11}^{(0)}\right)^* R_{1j}^{(0)}$$

R_{11}^{0} R_{11}^{0} R_{11}^{0} R_{11}^{0}

	By direct substitution	Simplified
$R_{11}^{(1)}$	$\epsilon + 1 + (\epsilon + 1)(\epsilon + 1)^*(\epsilon + 1)$	1^*
$R_{12}^{(1)}$	$0 + (\epsilon + 1)(\epsilon + 1)^*0$	1^*0
$R_{21}^{(1)}$	$\emptyset + \emptyset(\epsilon + 1)^*(\epsilon + 1)$	$\emptyset$
$R_{22}^{(1)}$	$\epsilon + 0 + 1 + \emptyset(\epsilon + 1)^*0$	$\epsilon + 0 + 1$

	Simplified
$R_{11}^{(1)}$	1^*
$R_{12}^{(1)}$	1^*0
$R_{21}^{(1)}$	$\emptyset$
$R_{22}^{(1)}$	$\epsilon + 0 + 1$

$RR^* = \{R, RR, RRR, \dots\}$.

$RR^* \neq R^*$

minimal R — minimal ada ε

$$R_{ij}^{(2)} = R_{ij}^{(1)} + R_{i2}^{(1)}\left(R_{22}^{(1)}\right)^* R_{2j}^{(1)}$$

$k = 2$.

	By direct substitution
$R_{11}^{(2)}$	$1^* + 1^*0(\epsilon + 0 + 1)^*\emptyset$
$R_{12}^{(2)}$	$\boxed{1^*0 + 1^*0(\epsilon + 0 + 1)^*(\epsilon + 0 + 1)}$ $= L(A)$
$R_{21}^{(2)}$	$\emptyset + (\epsilon + 0 + 1)(\epsilon + 0 + 1)^*\emptyset$
$R_{22}^{(2)}$	$\epsilon + 0 + 1 + (\epsilon + 0 + 1)(\epsilon + 0 + 1)^*(\epsilon + 0 + 1)$

$(\cancel{\varepsilon}+0+1)^* = \underbrace{(0+1)}_{\varepsilon}{}^*$

$R_{1j}^{(k)} = R_{14}^{(k)} \cup R_{15}^{(k)}$ Mis $F = \{q_4, q_5\}$.

$j \in F$

$1^*0 + 1^*0(\varepsilon+0+1)^*$

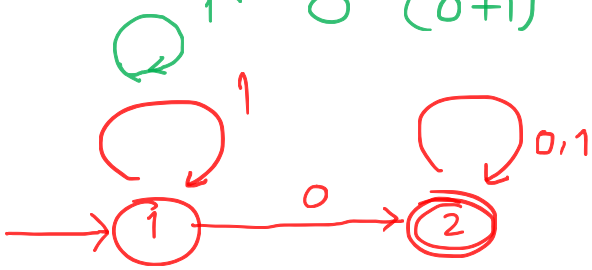
$\cancel{1^*0} + 1^*0\underbrace{(0+1)^*}_{\varepsilon}$

$1^*0(0+1)^*$

	By direct substitution
$R_{11}^{(2)}$	$1^* + 1^*0(\epsilon+0+1)^*\emptyset$
$R_{12}^{(2)}$	$1^*0 + 1^*0(\epsilon+0+1)^*(\epsilon+0+1)$
$R_{21}^{(2)}$	$\emptyset + (\epsilon+0+1)(\epsilon+0+1)^*\emptyset$
$R_{22}^{(2)}$	$\epsilon+0+1+(\epsilon+0+1)(\epsilon+0+1)^*(\epsilon+0+1)$

	Simplified
$R_{11}^{(2)}$	1^*
$R_{12}^{(2)}$	$1^*0(0+1)^*$
$R_{21}^{(2)}$	$\emptyset$
$R_{22}^{(2)}$	$(0+1)^*$

$1^* \quad 0 \quad (0+1)^*$.



$L(A) = R_{12}^{(2)} = 1^*0(0+1)^*$.

The final regex for A is

$$\boxed{R_{12}^{(2)} = 1^*0(0+1)^*}$$

Observations

There are n^3 expressions $R_{ij}^{(k)}$

Each inductive step grows the expression 4-fold

$R_{ij}^{(n)}$ could have size $\boxed{4^n}$

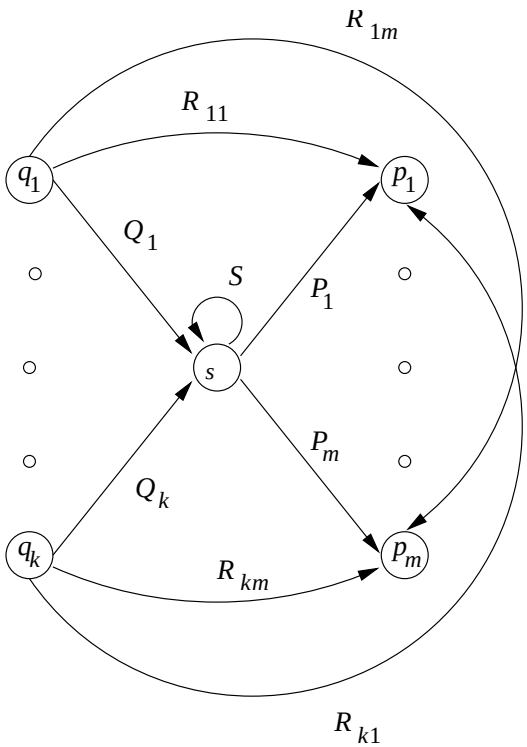
$R_{ij}^{k-1} + R_{ik}^{k-1} R_{kk}$

For all $\{i, j\} \subseteq \{1, \ldots, n\}$, $R_{ij}^{(k)}$ uses $R_{kk}^{(k-1)}$
so we have to write n^2 times the regex $R_{kk}^{(k-1)}$

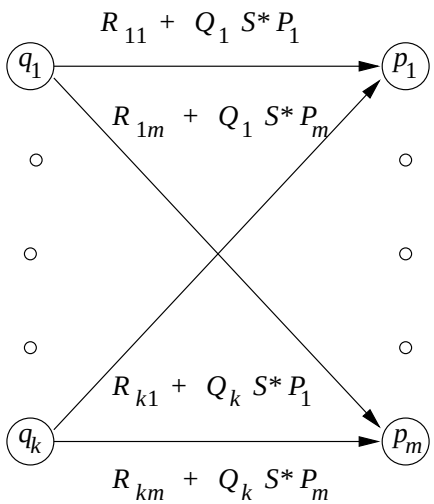
We need a more efficient approach:
the state elimination technique

The state elimination technique

Let's label the edges with regex's instead of symbols

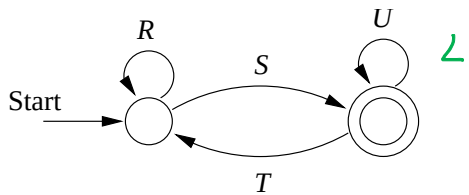


Now, let's eliminate state s.



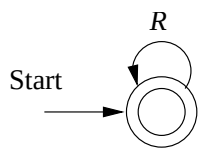
For each accepting state q eliminate from the original automaton all states exept q_0 and q.

For each $q \in F$ we'll be left with an A_q that looks like



that corresponds to the regex $E_q = (R+SU^*T)^*SU^*$

or with A_q looking like



$(R + R^*)^*$

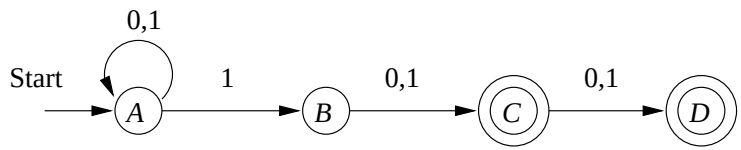
$RR^* = \{R, RR.$

corresponding to the regex $E_q = R^*$

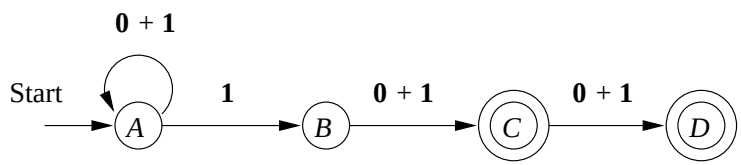
- The final expression is

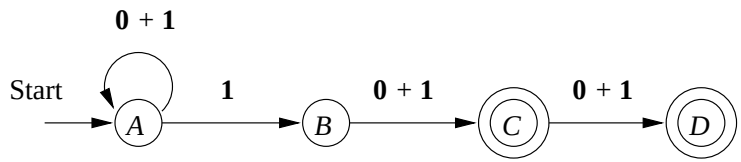
$$\bigoplus_{q \in F} E_q$$

Example: $\mathcal{A}$, where $L(\mathcal{A}) = \{W : w = x1b, \text{ or } w = x1bc, \ \ x \in \{0,1\}^*, \{b,c\} \subseteq \{0,1\}\}$

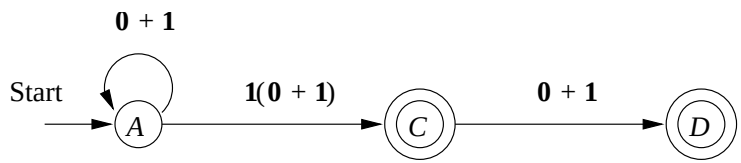


We turn this into an automaton with regex labels

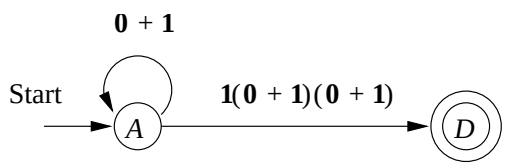




Let's eliminate state B

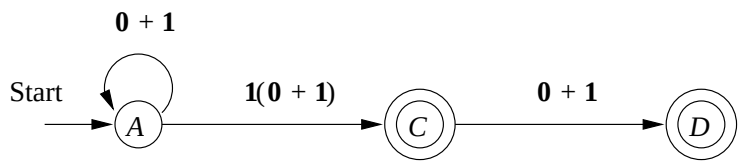


Then we eliminate state C and obtain $\mathcal{A}_D$

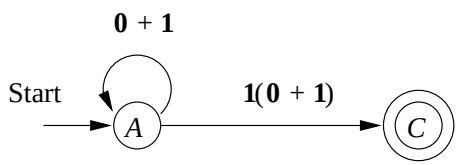


with regex $(\mathbf{0}+\mathbf{1})^*\mathbf{1}(\mathbf{0}+\mathbf{1})(\mathbf{0}+\mathbf{1})$

From



we can eliminate D to obtain $\mathcal{A}_C$



with regex $(\mathbf{0}+\mathbf{1})^*\mathbf{1}(\mathbf{0}+\mathbf{1})$

- The final expression is the sum of the previous two regex's:

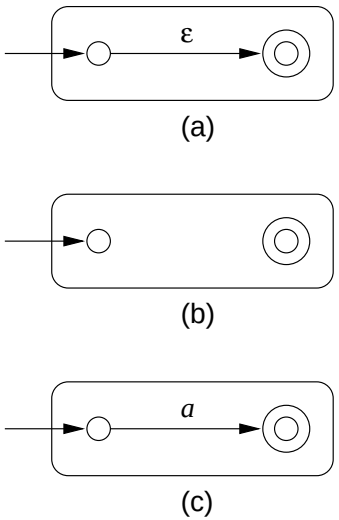
$$(\mathbf{0}+\mathbf{1})^*\mathbf{1}(\mathbf{0}+\mathbf{1})(\mathbf{0}+\mathbf{1}) + (\mathbf{0}+\mathbf{1})^*\mathbf{1}(\mathbf{0}+\mathbf{1})$$

From regex's to ϵ-NFA's

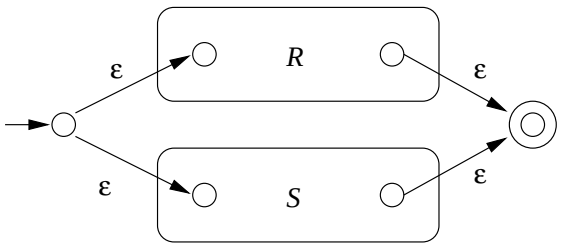
Theorem 3.7: For every regex R we can construct and ϵ-NFA A, s.t. $L(A) = L(R)$.

Proof: By structural induction:

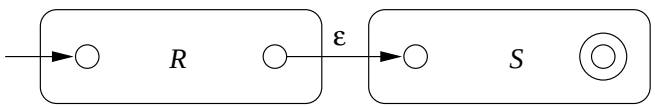
Basis: Automata for ϵ, $\emptyset$, and a.



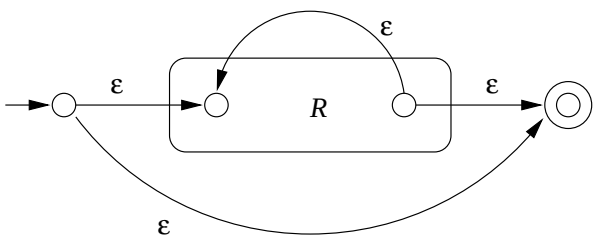
Induction: Automata for $R + S$, RS, and R^*



(a)

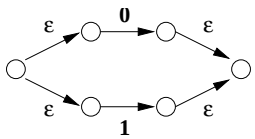


(b)

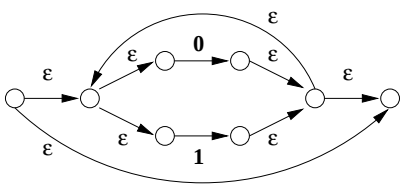


(c)

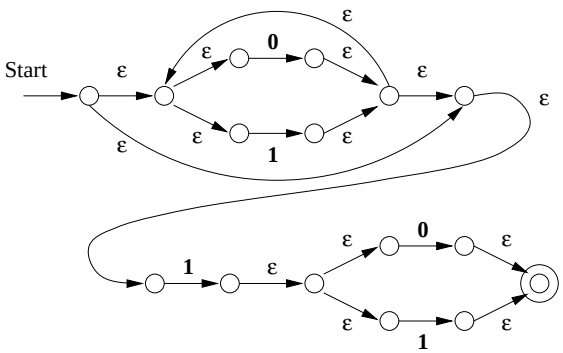
Example: We convert $(\mathbf{0}+\mathbf{1})^*\mathbf{1}(\mathbf{0}+\mathbf{1})$



(a)



(b)



(c)